

C266

The Ocean Systems

Armored Drill

8 concepts · 12 items · 4 transfer items

What this is. Every concept in this course's public description, re-asked with more disguise than a practice test uses. Each stem carries facts that do not decide it. Every wrong answer is a named trap.

Source. Scope from the course's public description in the WGU Institutional Catalog (September 2026); items written from the field's standard body of knowledge. No WGU assessment item, course material or proprietary content is used or reproduced. Not affiliated with or endorsed by Western Governors University.

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How to run it

Paper first. Pencil. One pass, no notes, a timer if you want the pressure.

1. **Before you read the options**, fill the strip under the item: the target word (what is actually being asked), and the one fact in the stem that decides it.
2. **Principle before arithmetic**. Name the rule, then compute.
3. **Camouflage**. Most of every stem does not matter. Ask: which single fact decides this?
4. **Inverse pairs are the bait**. If two options are opposites, one of them is the trap built for you.
5. **Mark sure, unsure or guess** honestly. Sure-and-wrong is the most valuable miss on the page — it is a wrong mental model.
6. **Absolutes** (always, never, guarantee) are usually wrong. And recognizing an answer is not the same as owning it.
7. Items with a **heavy black bar** on the left are **transfer items**: the same principle, reversed or solved for a different variable.
8. Stop at the STOP page. Then grade, fill the triage grid, and tally the traps you fell for. The biggest tally is what you drill next.

The strip, filled in, looks like this:

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	_____	

Items

1.

A coastal town known for its seafood festival has milder winters and cooler summers than an inland city at the same latitude. What best explains the difference?

- A. Water's high specific heat, from hydrogen bonding, makes the ocean slow to warm and cool.
- B. Coastal air always holds less carbon dioxide.
- C. Dissolved salt keeps seawater warm all year.
- D. Seawater is denser than fresh water.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

2.

Researchers aboard a ship from a university that has run ocean vessels since 1960 sample four water masses. Which is densest and most likely to sink?

- A. 25 °C, salinity 37.0
- B. 25 °C, salinity 35.0
- C. 2 °C, salinity 34.0
- D. 2 °C, salinity 34.9

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

3. [TRANSFER — reversed stem / different variable]

Melting Greenland ice adds fresh water to the North Atlantic where deep water forms. What is the expected effect on deep-water formation?

- A. It weakens, because fresher surface water is less dense and sinks less readily.
- B. All ocean currents stop worldwide at once.
- C. No effect, because meltwater is cold.
- D. It strengthens, because more water enters the ocean.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

4.

A sailor from a family-run charter company plans a route around the North Atlantic. In which direction does the main North Atlantic surface gyre circulate, and why?

- A. It is driven mainly by density differences.
- B. Counterclockwise, because of the Coriolis effect.
- C. Clockwise, driven by trade winds and westerlies deflected to the right by the Coriolis effect.
- D. It reverses twice a day with the tides.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

5. [TRANSFER — reversed stem / different variable]

In the Southern Hemisphere, in what direction does net Ekman transport move surface-layer water relative to the wind?

- A. 90° to the right of the wind
- B. About 45° to the left of the wind
- C. 90° to the left of the wind
- D. The same direction as the wind

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

6.

A research submersible whose dives are streamed live to classrooms collects basalt on both sides of the Mid-Atlantic Ridge. How does the age of the crust change moving away from the ridge axis?

- A. It is the same age everywhere.
- B. The oldest rock is at the ridge, since ridges are ancient mountains.
- C. It gets younger.
- D. It gets older on both sides, in a mirror-image pattern.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

7. [TRANSFER — reversed stem / different variable]

Where is the oldest oceanic crust most likely found?

- A. Anywhere, since oceanic crust is over 4 billion years old
- B. At hydrothermal vents
- C. Near deep-sea trenches far from spreading ridges
- D. At the crest of a mid-ocean ridge

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

8.

Off the coast of Peru, where fishing boats have been painted bright blue for generations, winds blow along the shore and push surface water offshore. Why are the waters so productive?

- A. The region gets more sunlight than anywhere else.
- B. Rivers deliver all of the nutrients.
- C. Warm surface water holds the most nutrients.
- D. Cold, nutrient-rich deep water upwells into the sunlit surface layer.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

9.

A marine ecology class at a school with a large saltwater aquarium models a food chain: phytoplankton → zooplankton → small fish → tuna. Phytoplankton capture 50,000 kcal/m² per year. About how much reaches the tuna?

- A. 5,000 kcal/m²
- B. 50 kcal/m²
- C. 50,000 kcal/m², since energy is conserved up the chain.
- D. 500 kcal/m²

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

10. [TRANSFER — reversed stem / different variable]

Seals eat fish, fish eat zooplankton, and zooplankton eat phytoplankton. If the seals need 200 kcal, about how much phytoplankton energy supports them?

- A. 200,000 kcal
- B. 2,000 kcal
- C. 20 kcal
- D. 20,000 kcal

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

11.

An oyster farm that sells to 30 restaurants reports thin, weak shells on its larvae. Local seawater pH has fallen from 8.2 to 8.0. What is the most likely mechanism?

- A.** More CO₂ supplies extra carbonate, so shells should thicken.
- B.** Absorbed CO₂ forms carbonic acid, lowering pH and reducing the carbonate ions larvae need to build shells.
- C.** The water is now acidic, since pH 8.0 is an acid.
- D.** Warmer water directly dissolves the shells.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

12.

A museum exhibit funded by a local bank explains how Earth's oceans and oxygen-rich air formed. Which sequence is best supported?

- A.** Oceans formed only from melting polar ice.
- B.** Volcanic water vapor (plus some from comets and asteroids) condensed into oceans; later, ocean cyanobacteria added oxygen to the air.
- C.** The early air was rich in oxygen, which combined with hydrogen to fill the oceans.
- D.** Land plants produced the first oxygen.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

STOP

Answers and dissection follow. Finish every item and every strip first.

Key and dissection

Struck-through text is the camouflage: facts that do not decide the item.

1. Answer A Concept 1

A coastal town ~~known for its seafood festival~~ has milder winters and cooler summers than an inland city at the same latitude. What best explains the difference?

Target word: cause of the moderated climate **Decides:** Water stores and releases heat slowly.

A	Correct	The ocean buffers temperature swings near the coast.
B	Absolute language	Not true, and not the cause.
C	Confusion set	Salt lowers the freezing point; it does not moderate climate.
D	Adjacent category	Density affects circulation, not heat storage.

2. Answer D Concept 2

Researchers aboard a ship ~~from a university that has run ocean vessels since 1960~~ sample four water masses. Which is densest and most likely to sink?

Target word: densest water **Decides:** Cold and salty together.

A	Dropped element	Considers salinity alone; the warmth makes it much lighter.
B	Inverted operation	Warm and moderately salty: among the least dense.
C	Confusion set	Cold but fresher than the correct choice.
D	Correct	Cold and salty gives the highest density.

3. Answer A Concept 2 · transfer

Melting Greenland ice adds fresh water to the North Atlantic where deep water forms. What is the expected effect on deep-water formation?

Target word: reverse: change input, predict circulation **Decides:** Fresher surface water is less dense.

A	Correct	Lower salinity reduces density.
B	Absolute language	Wind-driven currents continue; overturning slows.
C	Dropped element	Drops the salinity drop, which outweighs the cold here.
D	Inverted operation	Volume is not density.

4. Answer C Concept 3

A sailor ~~from a family-run charter company~~ plans a route around the North Atlantic. In which direction does the main North Atlantic surface gyre circulate, and why?

Target word: gyre direction **Decides:** Winds plus rightward Coriolis deflection in the Northern Hemisphere.

A	Confusion set	Density drives deep circulation; wind drives surface gyres.
B	Inverted operation	That is the Southern Hemisphere pattern.
C	Correct	Northern Hemisphere gyres turn clockwise.
D	Adjacent category	Tides are a different motion from gyres.

5. Answer C Concept 3 • transfer

In the Southern Hemisphere, in what direction does net Ekman transport move surface-layer water relative to the wind?

Target word: Ekman direction in the other hemisphere **Decides:** Deflection is to the left in the south; net transport is 90°.

A	Inverted operation	That is the Northern Hemisphere.
B	Confusion set	That is the surface current, not the net transport of the layer.
C	Correct	Southern Hemisphere deflection is leftward; the net is 90°.
D	Dropped element	Drops the Coriolis deflection.

6. Answer D Concept 4

A research submersible ~~whose dives are streamed live to classrooms~~ collects basalt on both sides of the Mid-Atlantic Ridge. How does the age of the crust change moving away from the ridge axis?

Target word: age pattern **Decides:** New crust forms at the ridge and moves outward.

A	Dropped element	Drops the spreading process.
B	Confusion set	Confuses ridges with old continental ranges.
C	Inverted operation	Reverses the spreading direction.
D	Correct	Seafloor spreading adds crust at the axis.

7. Answer C Concept 4 • transfer

Where is the oldest oceanic crust most likely found?

Target word: reverse: from age to location **Decides:** Crust is oldest farthest from ridges, near where it subducts.

A	Confusion set	Oceanic crust is recycled; the oldest is roughly 200 million years.
B	Adjacent category	Vents cluster at young, active ridges.
C	Correct	Old, dense crust sinks at subduction zones.
D	Inverted operation	That is the youngest crust.

8. Answer D Concept 5

Off the coast of Peru, ~~where fishing boats have been painted bright blue for generations~~, winds blow along the shore and push surface water offshore. Why are the waters so productive?

Target word: cause of productivity **Decides:** Offshore movement draws up deep water.

A	Confusion set	Light is common in the tropics; nutrients are the limit.
B	Adjacent category	Rivers add some, but upwelling drives this system.
C	Inverted operation	Warm surface water is usually nutrient-poor.
D	Correct	Nutrients plus light fuel phytoplankton.

9. Answer B Concept 6

A marine ecology class at a school ~~with a large saltwater aquarium~~ models a food chain: phytoplankton → zooplankton → small fish → tuna. Phytoplankton capture 50,000 kcal/m² per year. About how much reaches the tuna?

Target word: energy at the third consumer level **Decides:** Three transfers at 10% each.

A	Dropped element	Applies only one transfer.
B	Correct	50,000 → 5,000 → 500 → 50.
C	Absolute language	Most energy leaves each level as heat.
D	Confusion set	Stops at the small fish.

10. Answer A Concept 6 · transfer

Seals eat fish, fish eat zooplankton, and zooplankton eat phytoplankton. If the seals need 200 kcal, about how much phytoplankton energy supports them?

Target word: solve backward for producer energy **Decides:** Multiply by 10 for each of three steps down.

A	Correct	200 → 2,000 → 20,000 → 200,000.
B	Dropped element	Accounts for only one step.
C	Inverted operation	Divides instead of multiplying going down the chain.
D	Confusion set	Stops one level short, at zooplankton.

11. Answer B Concept 7

An oyster farm ~~that sells to 30 restaurants~~ reports thin, weak shells on its larvae. Local seawater pH has fallen from 8.2 to 8.0. What is the most likely mechanism?

Target word: acidification mechanism **Decides:** Absorbed CO₂ lowers pH and carbonate availability.

A	Inverted operation	Added CO ₂ lowers carbonate ion concentration.
B	Correct	Lower carbonate saturation makes calcification harder.
C	Confusion set	pH 8.0 is still basic; it is less basic.
D	Adjacent category	Warming is a separate stress; the pH drop points to CO ₂ chemistry.

12. Answer B Concept 8

A museum exhibit ~~funded by a local bank~~ explains how Earth's oceans and oxygen-rich air formed. Which sequence is best supported?

Target word: early Earth order **Decides:** Oceans first, oxygen from ocean microbes later.

A	Confusion set	Ice caps came after oceans existed.
B	Correct	Outgassing and cooling formed oceans before photosynthesis oxygenated the atmosphere.
C	Inverted operation	Oxygen came much later, from life.
D	Adjacent category	Ocean microbes did, long before land plants.

Triage grid

Write each item number in the box it belongs to.

	Right	Wrong
Sure	Owned.	Concept inversion — top priority. A wrong mental model; it survives rewording.
Unsure / guess	Recognition, not mastery. Re-drill with the transfer.	Not learned yet.

Trap tally

For every miss, find the letter you chose in the key and tally its trap. The biggest number is the pattern to drill.

Trap	What it does	In this drill	Your misses
Dropped element	Leaves out a required term or step.	6	
Wrong stem number	A figure copied from the prompt that is not the answer.	0	
Adjacent category	The right idea in the wrong bucket.	6	
Inverted operation	Direction flipped: add/subtract, cause/effect, higher/lower.	10	
Absolute language	Always, never, guarantee — where judgment applies.	3	
Confusion set	Every option is real; the test is exact role-matching.	11	
Camouflage	Answers a stem fact that was never the question.	0	

Anchor rows — the concept list

One row per concept: the rule to lock in, the trap that catches people, and the tell in the question that signals it.

#	Concept	Rule	Trap	Tell
1	Water's high heat capacity	Hydrogen bonding gives water a high specific heat, so the ocean warms and cools slowly and moderates coastal climate.	Crediting salt or density for the moderating effect.	A coastal place with milder seasons than an inland one.
2	Seawater density and thermohaline circulation	Colder and saltier water is denser and sinks; deep-water formation at high latitudes drives global overturning circulation.	Judging density by salinity alone or temperature alone.	Water masses listed with temperature and salinity.
3	Wind-driven gyres and the Coriolis effect	Winds plus Coriolis deflection (right in the Northern Hemisphere, left in the Southern) create gyres; net Ekman transport is 90° from the wind.	Reversing hemispheres or attributing surface gyres to tides.	Direction of a basin-wide current or wind-driven water movement.
4	Seafloor spreading	New oceanic crust forms at mid-ocean ridges, so crust gets older with distance from the ridge and is oldest near subduction zones.	Thinking ridges are old mountains with the oldest rock.	Rock ages sampled across a ridge.
5	Upwelling and productivity	When surface water is pushed offshore, cold, nutrient-rich deep water rises and fuels high plankton and fish productivity.	Thinking warm surface water is the most nutrient-rich.	Alongshore winds and an unusually rich fishery.
6	Energy flow in marine food webs	Only about 10% of energy passes from one trophic level to the next.	Assuming energy passes up the chain intact or miscounting steps.	Producer energy and a consumer several levels up.
7	Ocean acidification	The ocean absorbs CO ₂ , forming carbonic acid that lowers pH and reduces carbonate ions that shell-builders need.	Thinking extra CO ₂ supplies more carbonate, or that pH 8.0 is acidic.	Falling pH and weak shells.
8	Origin of the oceans and early life	Oceans formed as water vapor from volcanic outgassing (plus some delivered by comets and asteroids) condensed on a cooling Earth; ocean cyanobacteria later oxygenated the atmosphere.	Thinking oxygen or land plants came first.	A sequence of early Earth events.

Course scope (public description): This course investigates the complex ocean system by looking at the way its components—atmosphere, biosphere, geosphere, hydrosphere—interact. Specific topics include the origins of Earth's oceans and the early history of life; physical characteristics and geologic processes of the ocean floor; chemistry of the water molecule; energy flow between air and water and how ocean surface currents and deep circulation patterns affect weather and climate; marine biology and why ecosystems are an integral part of the ocean system; the effects of human activity; and the role of professional educators in teaching about ocean systems.